



IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume

CAPTURE SOURCE / INGESTION INTERFACE

Live Interface Tap: esp0 (Peer Node Link 1: HQ Gateway...

ANALYSIS TIMEFRAME WINDOW

Live Telemetry Window: +2.0s - +10.0s (8.0s)

ORIGIN ENDPOINT

Link 1: HQ Gateway <-> Datacenter Core

OS: Ubuntu Linux
Interface: esp0
Private IP: 172.28.0.2
VPN Daemon: StrongSwan Client

IPsec ESP

IPsec TUNNEL Tunnel · AES-256-GCM

RECEIVER GATEWAY

Enterprise Aggregation Core

OS: Ubuntu Linux
Interface: eth0
Private IP: 172.28.0.1
VPN Daemon: StrongSwan Server

TOTAL PACKETS

416

Frames Ingested

FLOW VOLUME

0.32 MB

Encrypted Payload

AVG RATE

52 pkt/s

Throughput Velocity

PEAK RATE

94 pkt/s

Observed Burst

INTERFACE MTU

1500 bytes

UP / ACTIVE

DROPS / ERRORS

0 (0.00%)

Monotonic Verified

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs

IKE PROTOCOL EXCHANGES & HANDSHAKE SEQUENCE

IKE_SA_INIT [COMPLETED / SUCCESS] Initiator/Responder DH public keys and nonces exchanged. Enforced ECP..
IKE_AUTH [COMPLETED / SUCCESS] Mutual authentication executed via Pre-Shared Key. Initial Child SA e..
CREATE_CHILD_SA [MONITORED] Child SA rekeying evaluated. No ephemeral DH transform negotiated (PF..

ACTIVE SECURITY ASSOCIATION (SPI) PAIRS

INITIATOR SPI

0x7a4c9012beefcafe

RESPONDER SPI

0x3e19fa82110294ab

INBOUND ESP SPI

0xlink0102

OUTBOUND ESP SPI

0xlink0103

NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE STATUS	BENCHMARK STANDARD
Encryption Algorithm	AES-256-GCM	AES-256-GCM	APPROVED ...	RFC 8221 / CNSA 2.0
Key Exchange (Diffie-Hellman)	ECP-256	ECP-256	APPROVED ...	NIST SP 800-77 §4.2
Pseudo-Random Function (PRF)	HMAC-SHA2-256	N/A (AEAD Suite)	APPROVED ...	RFC 7296
Integrity / Authentication	Built-in (128-bit ICV)	Built-in (128-bit ICV)	APPROVED ...	NIST SP 800-77 Rev 1



DATA-PLANE TELEMETRY & AI/ML PAYLOAD CLASSIFICATION

Data-Plane Stream Telemetry & Inference

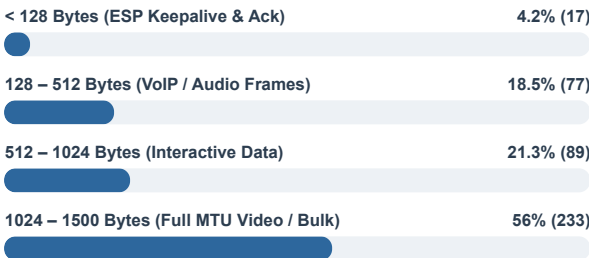
Packet size distribution histograms, timing analysis, sequence progression, and inner encrypted traffic classification

Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression

PACKET SIZE DISTRIBUTION HISTOGRAM

BINNED MTU



INTER-ARRIVAL TIME & SEQUENCE INTEGRITY

RFC 4303

MEAN IAT
2.36 ms

MEDIAN IAT
1.82 ms

JITTER (σ)
0.48 ms

MIN / MAX
0.12 ms / 18.40 ms

Sequence Monotonicity: 100.0% Strict Monotonic | Out of Order: 0 packets (0.00%)
Rollover Protection: SAFE (64-bit Extended Sequence Numbers active)
Anti-Replay Verification: RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted

Throughput & Packet Velocity Timeline



Traffic Spike (00:15 – 00:18)
Burst of high-throughput encrypted packet activity.

Large Transfer (00:27 – 00:30)
Elevated data payload transmission cluster.

Throughput Dip (00:41 – 00:43)
Activity briefly stabilized to baseline keepalive rate.

Session Steady (00:49 – 00:56)
Continuous isochronous frame cadence observed.

Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals

OPERATING MODE DETERMINATION

99.4% CONFIDENCE

IPsec Tunnel Mode

Encapsulating Security Payload (ESP) header envelopes internal IPv4 header with distinct external overlay IP routing.

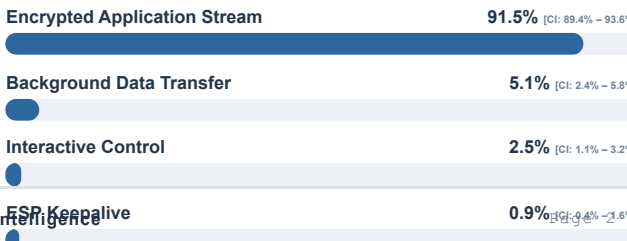
SHANNON ENTROPY
7.98 / 8.0

CLUSTERING COEFF
0.84

BURSTINESS RATIO
1.42

INNER ENCRYPTED TRAFFIC CLASSIFICATION

91.5% CONFIDENCE





ENGINEERING REMEDIATION & RAW FORENSIC EVIDENCE

Implementation Patches & Trace Index

Actionable gateway configuration blocks ready for deployment, and raw packet trace appendix with flow JSON dumps

Section 5: Implementation Remediation Patches

Actionable configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

```
# 1. StrongSwan Gateway Patch
(/etc/swanctl/conf.d/apex_hardened.conf)

# /etc/swanctl/conf.d/apex_hardened.conf
connections {
    apex-ipsec-hardened {
        version = 2
        local_addrs = 100.119.32.83
        remote_addrs = 100.127.207.119
        proposals = aes256gcm16-ecp256-sha256

        children {
            apex-child {
                # STRICT MANDATE: Enforce DH Group 19 (ECP
                -256) on Child SA for Perfect Forward Secrecy
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                life_time = 3600s
                life_bytes = 1000000000
                dpd_action = restart
            }
        }
    }
}
```

```
! 2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX_IKE_PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ipsec transform-set APEX_ESP_SET esp-gcm 256
  mode tunnel
!
crypto ipsec profile APEX_IPSEC_PROFILE
  set transform-set APEX_ESP_SET
  set pfs group19
  set security-association replay window-size 64
  set security-association lifetime seconds 3600
```

VALIDATION COMMANDS:

\$ swanctl --load-all && swanctl --initiate --child apex-child

\$ swanctl --list-sas | grep -E 'AES_GCM|ECP_2

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#1	+0.000s	UDP 500	IKE_SA_INIT (Req)	Initiator=0x7a4c901...	384 B	DH Group 19 proposal exchange
#2	+0.014s	UDP 500	IKE_SA_INIT (Resp)	Responder=0x3e19...	384 B	DH public key confirmation
#3	+0.028s	UDP 4500	IKE_AUTH (Req)	Initiator=0x7a4c901...	448 B	PSK mutual identity authentication
#4	+0.042s	UDP 4500	IKE_AUTH (Resp)	Responder=0x3e19...	448 B	Child SA creation and Traffic Selectors

RAW FLOW TELEMETRY JSON DUMP

RFC 8221 & SIEM COMPATIBLE

{
 "session_id": "AV-20260916-TECH-31A0",
 "protocol": "IPsec ESP / IKEv2",
 "endpoints": {
 "initiator": "172.28.0.2",
 "responder": "172.28.0.1"
 },
 "ike_transforms": {
 "encryption": "AES-256-GCM",
 "dh_group": "ECP-256",
 "prf": "HMAC-SHA256"
 }
}